Cubical Hyperresolutions and Mixed Hodge Structures

A seminar handout with four worked examples

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1 Conventions and the question to keep in view

All varieties are reduced complex algebraic varieties. Throughout,

$$H^k(X) = H^k(X(\mathbb{C}), \mathbb{Q})$$

means *ordinary singular cohomology*, not cohomology with compact supports. A point has $H^0 = \mathbb{Q}(0)$, and $H^2(\mathbb{P}^1) = \mathbb{Q}(-1)$: our Tate convention makes $\mathbb{Q}(-1)$ pure of weight 2 and Hodge type $(1,1)$ ([1, Examples 2.2(ii), 3.2(3)]).

A *rational mixed Hodge structure* (MHS) on a finite-dimensional $\mathbb{Q}$-vector space H consists of an increasing filtration W on H and a decreasing filtration F on $H_{\mathbb{C}} = H \otimes_{\mathbb{Q}} \mathbb{C}$, such that each $\mathrm{Gr}^W_m H$ is a pure Hodge structure of weight m. Concretely,

$$(\mathrm{Gr}^W_m H)_{\mathbb{C}} = \bigoplus_{p+q=m} H^{p,q}_m, \qquad \overline{H^{p,q}_m} = H^{q,p}_m, \qquad F^a(\mathrm{Gr}^W_m H)_{\mathbb{C}} = \bigoplus_{p \geq a} H^{p,m-p}_m.$$

The *degreewise mixed Hodge numbers* are ([1, §3.1, p. 62])

$$\boxed{h^{p,q}(H^k(X)) = \dim_{\mathbb{C}} \mathrm{Gr}^p_F \, \mathrm{Gr}^W_{p+q} \, H^k(X)_{\mathbb{C}}.}$$

For smooth projective X, $H^k(X)$ is pure of weight k. For singular or open varieties, several weights can occur. But singularity alone does not force a cohomology group to have several weights: our first two singular curves already illustrate this distinction.

The guiding question. How can smooth pieces recover both the topology of a singular variety and its Hodge filtrations? A resolution supplies smooth pieces; a hyperresolution also records how they are glued. The same diagram produces a complex computing H^k, and a filtered complex defining its MHS. The latter contains information that the homotopy type alone cannot determine.

2 How to build a cubical hyperresolution

Following [1, Definition 5.1], an N-cubical variety has entries X_I indexed by $I \subseteq \{0, \dots, N-1\}$ and compatible maps

$$d_{IJ} : X_J \longrightarrow X_I \quad (I \subseteq J), \qquad X_\varnothing = X.$$

Its associated augmented semi-simplicial variety has

$$X_r = \coprod_{|I|=r+1} X_I \quad (r \geq 0), \qquad \varepsilon : X_\bullet \longrightarrow X \qquad ([1, \text{Remark } 5.4(2)])$$

with face maps obtained by deleting one index. Thus a square is a 2-cubical diagram; its associated semi-simplicial object has two levels.

The augmentation is of *cohomological descent* if

$$\mathbb{Q}_X \xrightarrow{\sim} R\varepsilon_* \mathbb{Q}_{X_\bullet}$$

is a quasi-isomorphism. A *cubical hyperresolution* has this property, smooth nonempty-index entries, and proper structural maps to X. For a pair (X, D), the inverse image of D on every irreducible component must be empty, the whole component, or a simple normal crossings divisor ([1, Definitions 5.6, 5.8, 5.10]). Properness here is relative to X; the entries are proper over $\mathbb{C}$ when X is proper.

The basic construction step: the discriminant square. Start with a proper resolution $\nu : Y \to X$. Choose its discriminant $S \subseteq X$, so that ν is an isomorphism off S, and put $E = (\nu^{-1}S)_{\mathrm{red}}$. The square is

$$\begin{array}{ccc} E = X_{\{0,1\}} & \xrightarrow{\quad j \quad} & Y = X_{\{0\}} \\ {\scriptstyle q}\downarrow & & \downarrow{\scriptstyle \nu} \\ S = X_{\{1\}} & \xrightarrow[\quad i \quad]{} & X = X_\varnothing \end{array}$$

This square is of cohomological descent ([1, Lemma–Definition 5.17]). It is already a hyperresolution if Y, S, E are smooth. Otherwise resolve the remaining singular corners *compatibly with their maps*, form further discriminant squares, and assemble them into a higher cube. For an open variety, do this for a compact pair, arranging simple normal crossings boundaries. Existence is [1, Theorem 5.26], read with the author's corrections [2]; no dimension estimate from its uncorrected statement is needed here.

Curves are especially concrete. If X is a reduced curve, take Y to be its normalization and S its finite singular locus. Then E is a finite reduced set, so every nonempty corner is smooth and the construction is finished ([1, Example 5.12]). Normalization alone forgets which preimages must be identified. The corners S and E, together with their maps, restore precisely this gluing data.

3 Descent turns the diagram into a cohomology calculation

For a normalization square of a curve, finiteness gives the exact sequence of sheaves on X

$$0 \longrightarrow \mathbb{Q}_X \longrightarrow \nu_*\mathbb{Q}_Y \oplus i_*\mathbb{Q}_S \xrightarrow{j^*-q^*} (i\circ q)_*\mathbb{Q}_E \longrightarrow 0. \tag{1}$$

At a singular point with b branches the last map is

$$\mathbb{Q}^b \oplus \mathbb{Q} \longrightarrow \mathbb{Q}^b, \qquad (a_1,\dots,a_b,c) \longmapsto (a_1-c,\dots,a_b-c).$$

It is surjective, with the one-dimensional diagonal kernel required by the constant sheaf. Away from S, exactness is immediate. This verifies descent directly in our curve examples. For general proper modifications, one uses the derived statement below, rather than asserting (1).

The cone and the long exact sequence. Cohomological descent gives

$$R\Gamma(X,\mathbb{Q}) \simeq \mathrm{Cone}\Big[R\Gamma(Y,\mathbb{Q}) \oplus R\Gamma(S,\mathbb{Q}) \xrightarrow{\delta} R\Gamma(E,\mathbb{Q})\Big][-1], \qquad \delta = j^*-q^*. \tag{2}$$

Writing $A = R\Gamma(Y,\mathbb{Q}) \oplus R\Gamma(S,\mathbb{Q})$ and $B = R\Gamma(E,\mathbb{Q})$, the shifted cone has degree-n term $A^n \oplus B^{n-1}$ and differential $(a,b) \mapsto (d_A a, \delta a - d_B b)$. Thus degree-zero data on E enters total degree one: its cokernel can create $H^1(X)$. Consequently,

$$\begin{aligned} \cdots \longrightarrow H^k(X) &\longrightarrow H^k(Y) \oplus H^k(S) \xrightarrow{\delta} H^k(E) \\ &\longrightarrow H^{k+1}(X) \longrightarrow \cdots . \end{aligned}$$

This is an exact sequence of MHS, not just of vector spaces ([1, Theorem 5.35 and Corollary–Definition 5.37]). Changing all signs in δ does not change its kernels and cokernels.

The larger cube. The analogous total complex has the spectral sequence

$$\boxed{E_1^{r,s} = H^s(X_r,\mathbb{Q}) \implies H^{r+s}(X,\mathbb{Q}), \qquad d_1 = \sum_j (-1)^j d_j^*.} \tag{3}$$

For indices $i_0 < \cdots < i_{r+1}$, the component of the differential is

$$(d_1\alpha)_{\{i_0,\dots,i_{r+1}\}} = \sum_{j=0}^{r+1} (-1)^j d^*_{\{i_0,\dots,\widehat{i_j},\dots,i_{r+1}\},\{i_0,\dots,i_{r+1}\}} \alpha_{\{i_0,\dots,\widehat{i_j},\dots,i_{r+1}\}}.$$

When all levels are smooth and proper. Each $E_1^{r,s}$ is pure of weight s, and d_1 preserves this weight. Morphisms of MHS are strict for both filtrations, so their kernels and cokernels carry the induced MHS; hence $E_2^{r,s}$ is pure of weight s. A higher differential $d_\ell : E_\ell^{r,s} \to E_\ell^{r+\ell,s-\ell+1}$, $\ell \geq 2$, would map between different pure weights and is therefore zero. Hence

$$E_2 = E_\infty, \qquad \boxed{\mathrm{Gr}_s^W H^n(X) \simeq E_2^{n-s,s}.} \tag{4}$$

In the examples, (3) supplies ordinary cohomology and (4) identifies its weights. For open levels the entries need not be pure; this argument alone does *not* give degeneration at E_2.

4 Where the mixed Hodge structure actually comes from

The weights in (4) are a consequence of a filtered construction. Here is that construction, including the indexing shift that matters in examples.

Step 1: compactify and resolve a pair. Given U, choose a proper variety $\overline{U}$ containing it as a dense open subset and put $T = \overline{U} \setminus U$. Choose a cubical hyperresolution $(\overline{X}_\bullet, D_\bullet)$ of $(\overline{U}, T)$. The pairs $(\overline{X}_r, D_r)$ are logarithmic pairs in the sense of [1, Definition 5.31]. On a component contained in D_r the relevant complex is zero; otherwise D_r is empty or a simple normal crossings divisor. Write $j_r : U_r = \overline{X}_r \setminus D_r \hookrightarrow \overline{X}_r$.

Step 2: put a mixed Hodge complex on each level. The complex $\mathrm{Hdg}^\bullet(\overline{X}_r \log D_r)$ has rational realization $R(j_r)_*\mathbb{Q}_{U_r}$ and complex realization $\Omega^\bullet_{\overline{X}_r}(\log D_r)$, connected by the logarithmic de Rham comparison. Its Hodge filtration is

$$F^a\Omega^\bullet(\log D_r) = \Omega^{\geq a}(\log D_r).$$

The level weight filtration on the complex realization counts logarithmic factors: locally, W_m is generated by forms with at most m factors dz_i/z_i in coordinates adapted to D_r. Its rational counterpart is matched by the filtered comparison. Residues explain why logarithmic terms introduce Tate twists ([1, Theorem 4.2, Lemma 4.6, and Proposition–Definition 4.11]).

Step 3: totalize, shifting the level weights. Take suitable functorial resolutions and derived global sections first, and denote the resulting level complexes by $K_r^\bullet$. With the alternating face differential, put

$$K^n = \mathrm{Tot}(K_\bullet)^n = \bigoplus_{r\geq 0} K_r^{n-r}.$$

Peters–Steenbrink's total filtrations are

$$\boxed{(W_mK)^n = \bigoplus_{r\geq 0} W_{m+r}K_r^{n-r}, \qquad (F^aK)^n = \bigoplus_{r\geq 0} F^aK_r^{n-r}.} \tag{5}$$

This is the convention of [1, Definition 5.32]. The rational total complex computes $H^*(U, \mathbb{Q})$ by descent. Its filtrations induce the MHS through

$$\boxed{W_mH^n(U) = \mathrm{im}\big[H^n(W_{m-n}K_{\mathbb{Q}}) \to H^n(K_{\mathbb{Q}})\big],} \tag{6}$$

with F induced on complex cohomology ([1, Theorem 3.18, II(i)]). The resulting MHS is independent of the compactification and hyperresolution, and algebraic pullbacks respect it ([1, Theorem 5.33]). This supplies the definition, not only its dimensions.

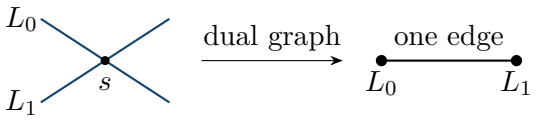
Reading the two shifts. For smooth proper levels with empty boundary, the level weight filtration starts at 0. Column r then enters total W at index $-r$ by (5), and contributes to weight $n - r = s$ in degree $n = r + s$ by (6). Thus incidence gluing can produce weights below the cohomological degree. Boundary logarithms can produce weights above it. Both effects will appear together in the punctured nodal curve.

5 Two projective curves: a tree and a loop

Example 1: two lines meeting once. Let $X = L_0 \cup L_1 = \{xy = 0\} \subset \mathbb{P}^2$. Its normalization square has

$$Y = \mathbb{P}^1 \sqcup \mathbb{P}^1, \qquad S = \{s\}, \qquad E = \{a, b\},$$

where a and b lie on different components of Y. The following picture is a schematic of incidence, not a picture of the real dimension of a complex curve.



The only gluing map in degree zero is

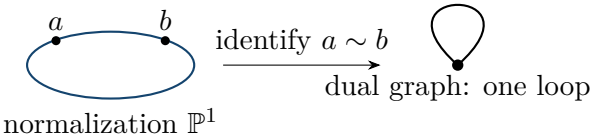
$$\delta : \mathbb{Q}^2 \oplus \mathbb{Q} \longrightarrow \mathbb{Q}^2, \qquad (a_0, a_1, c) \longmapsto (a_0 - c, a_1 - c).$$

It has rank 2 and diagonal kernel $\mathbb{Q}$. All positive-degree cohomology of S, E vanishes; $H^1(Y) = 0$ and $H^2(Y) = \mathbb{Q}(-1)^2$. The exact sequence (2) therefore gives, as MHS,

$$\boxed{H^0(X) = \mathbb{Q}(0), \qquad H^1(X) = 0, \qquad H^2(X) = \mathbb{Q}(-1)^2.}$$

The only nonzero mixed Hodge numbers are $h^{0,0}(H^0) = 1$ and $h^{1,1}(H^2) = 2$. This singular projective curve has pure cohomology of the usual weights.

Example 2: an irreducible rational nodal cubic. Let C be such a cubic and s its node. Now $Y = \mathbb{P}^1$, $S = \{s\}$, and $E = \{a, b\} \subset \mathbb{P}^1$ consists of two points on the *same* component.



The degree-zero map is

$$\delta : \mathbb{Q} \oplus \mathbb{Q} \longrightarrow \mathbb{Q}^2, \qquad (u, v) \longmapsto (u - v, u - v).$$

Its kernel and cokernel both have dimension 1. Hence

$$\boxed{H^0(C) = \mathbb{Q}(0), \qquad H^1(C) = \mathbb{Q}(0), \qquad H^2(C) = \mathbb{Q}(-1).}$$

In particular $h^{0,0}(H^1(C)) = 1$: the graph loop contributes weight 0, not weight 1 or 2. It is $E_2^{1,0}$ in (4). Topologically one may recognize $C \simeq S^2 \vee S^1$, but the weight-zero assertion comes from the algebraic descent diagram.

6 A surface: constructing a hyperresolution of $X \times X$

Take $X = L_0 \cup_s L_1$ from Example 1, with $L_i \simeq \mathbb{P}^1$, and put $Z = X \times X$, $t = (s, s)$. We now carry out the construction of Section 2: first resolve Z, then resolve the singular corners together with their maps. At t the local equations are $xy = uv = 0$ in $\mathbb{C}^4$. This is not a simple normal crossings surface: four surface components meet there.

6.1 The first discriminant square

The normalization separates the four surface components:

$$\nu : N = \coprod_{i,j=0,1} N_{ij} \longrightarrow Z, \qquad N_{ij} = L_i \times L_j \simeq \mathbb{P}^1 \times \mathbb{P}^1.$$

It is an isomorphism away from the four curves

$$S = A_0 \cup A_1 \cup B_0 \cup B_1, \qquad A_i = L_i \times \{s\}, \quad B_j = \{s\} \times L_j.$$

All four meet only at t. Write s_i for the point of L_i above s. In N_{ij} the reduced inverse image of S is the cross

$$E_{ij} = H_{ij} \cup V_{ij}, \qquad H_{ij} = L_i \times \{s_j\}, \quad V_{ij} = \{s_i\} \times L_j, \quad H_{ij} \cap V_{ij} = \{p_{ij}\},$$

where $p_{ij} = (s_i, s_j)$. Thus $E = \coprod_{i,j} E_{ij}$, and our first discriminant square is

$$\begin{array}{ccc} E & \longrightarrow & N \\ {\scriptstyle q}\downarrow & & \downarrow{\scriptstyle \nu} \\ S & \longrightarrow & Z \end{array} \qquad \begin{array}{l} H_{ij} \xrightarrow{\sim} A_i, \\ V_{ij} \xrightarrow{\sim} B_j, \\ p_{ij} \longmapsto t. \end{array}$$

It has cohomological descent by [1, Lemma–Definition 5.17], but it is *not yet a hyperresolution*: S and E are singular. Resolving S alone would leave the singular upper-left corner and its map untreated.

6.2 Resolve the singular face, then assemble the cube

We resolve the entire 1-cubical diagram $q : E \to S$ compatibly. Its normalization is the morphism

$$\widetilde{q} : \widetilde{E} = \coprod_{i,j} (H_{ij} \sqcup V_{ij}) \longrightarrow \widetilde{S} = A_0 \sqcup A_1 \sqcup B_0 \sqcup B_1, \qquad H_{ij} \mapsto A_i, \quad V_{ij} \mapsto B_j,$$

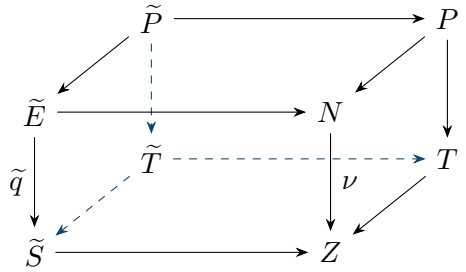
with each branch map an isomorphism. Denote the normalization maps by $b : \widetilde{E} \to E$ and $a : \widetilde{S} \to S$. Set $T = \{t\}$, $P = \{p_{ij}\}_{i,j}$, $\widetilde{T} = a^{-1}(T)$ and $\widetilde{P} = b^{-1}(P)$, always with reduced structures. There are $1, 4, 4, 8$ points in $T, P, \widetilde{T}, \widetilde{P}$, respectively. The two compatible discriminant squares are

$$\begin{array}{ccc} \widetilde{P} & \longrightarrow & \widetilde{E} \\ \downarrow & & \downarrow{\scriptstyle b} \\ P & \longrightarrow & E \end{array} \qquad \longrightarrow \qquad \begin{array}{ccc} \widetilde{T} & \longrightarrow & \widetilde{S} \\ \downarrow & & \downarrow{\scriptstyle a} \\ T & \longrightarrow & S \end{array}$$

The map between squares is q on E, $\widetilde{q}$ on $\widetilde{E}$, $p_{ij} \mapsto t$ on P, and the induced branch map $\widetilde{P} \to \widetilde{T}$. Every corner of these squares except E, S is smooth. This is precisely the second construction step in the proof of [1, Theorem 5.26, p. 122], with the corrections [2].

The resulting 3-cube. The old singular entries S, E are replaced by the normalization and point data. Index the resulting cube by subsets of $\{0,1,2\}$:

I	Z_I	Components
$\varnothing$	Z	the original surface
$\{0\}$	N	four copies of $\mathbb{P}^1 \times \mathbb{P}^1$
$\{1\}$	$\widetilde{S}$	four copies of $\mathbb{P}^1$
$\{2\}$	T	one point
$\{0,1\}$	$\widetilde{E}$	eight copies of $\mathbb{P}^1$
$\{0,2\}$	P	four points
$\{1,2\}$	$\widetilde{T}$	four points
$\{0,1,2\}$	$\widetilde{P}$	eight points



Here is every map not already specified by the first square and the normalizations. On $\widetilde{T}$, label the four points by a_i, b_j according to their branches A_i, B_j. On $\widetilde{P}$, label the two points over p_{ij} by h_{ij}, v_{ij}. Then

$$\begin{aligned}
&\widetilde{E} \to N: \quad H_{ij}, V_{ij} \hookrightarrow N_{ij}, && P \to N: \ p_{ij} \hookrightarrow N_{ij},\\
&\widetilde{T} \to \widetilde{S}: \quad a_i \in A_i, \ b_j \in B_j, && P \to T, \ \widetilde{T} \to T: \ \text{all points} \mapsto t,\\
&\widetilde{P} \to \widetilde{E}: \quad h_{ij} \in H_{ij}, \ v_{ij} \in V_{ij}, && \widetilde{P} \to P: \ h_{ij}, v_{ij} \mapsto p_{ij},\\
&\widetilde{P} \to \widetilde{T}: \quad h_{ij} \mapsto a_i, \ v_{ij} \mapsto b_j.
\end{aligned}$$

The singleton entries map to Z by ν, $\widetilde{S} \xrightarrow{a} S \hookrightarrow Z$, and $T \hookrightarrow Z$. All other maps are composites. For example, either route $h_{ij} \to N$ gives p_{ij}, and either route $h_{ij} \to \widetilde{S}$ gives a_i; the v_{ij} checks are identical. These branch labels verify that every face commutes.

All maps are finite, hence proper, and all nonempty-index entries are smooth and projective. The boundary is empty. The remaining issue is cohomological descent, checked next.

What has been assembled? The initial square and the two normalization squares are discriminant squares. The faces of the assembled cube need not themselves be discriminant squares. In particular, the face with entries $Z, N, \widetilde{S}, \widetilde{E}$ alone will leave a loop over t. The four point corners are essential to the full diagram.

6.3 Why this cube is a hyperresolution

Deleting one index gives the semi-simplicial levels

$$\begin{aligned} Z_0 &= N \sqcup \widetilde{S} \sqcup T = \coprod_4 (\mathbb{P}^1 \times \mathbb{P}^1) \;\sqcup\; \coprod_4 \mathbb{P}^1 \;\sqcup\; \mathrm{pt}, \\ Z_1 &= \widetilde{E} \sqcup P \sqcup \widetilde{T} = \coprod_8 \mathbb{P}^1 \;\sqcup\; \coprod_8 \mathrm{pt}, \qquad Z_2 = \widetilde{P} = \coprod_8 \mathrm{pt}. \end{aligned} \tag{7}$$

Let $\varepsilon_r : Z_r \to Z$ be the augmentation maps. They are finite, so proper base change identifies the stalks of $R^k(\varepsilon_r)_*\mathbb{Q}$ with the cohomology of finite fibres; these sheaves vanish for $k > 0$. Thus descent reduces to exactness of

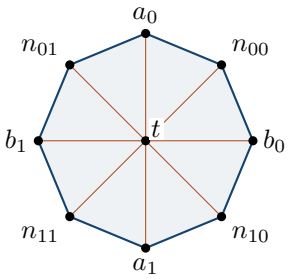
$$0 \longrightarrow \mathbb{Q}_Z \longrightarrow (\varepsilon_0)_*\mathbb{Q}_{Z_0} \longrightarrow (\varepsilon_1)_*\mathbb{Q}_{Z_1} \longrightarrow (\varepsilon_2)_*\mathbb{Q}_{Z_2} \longrightarrow 0,$$

with alternating face maps. At $z \in Z$ this is the augmented cochain complex of the finite simplicial complex whose vertices, edges and triangles are the fibres of Z_0, Z_1, Z_2. There are three cases.

Off S: one vertex. The normalization has one preimage and all other fibres are empty. The stalk complex is $0 \to \mathbb{Q} \to \mathbb{Q} \to 0$.

On $S \setminus T$: a tree. A point on $A_i \setminus \{t\}$ has two normalization preimages, one in each N_{i0}, N_{i1}, and one preimage in $\widetilde{S}$. The two H-branches give edges joining the latter vertex to the former two. Thus the fibre has three vertices and two edges. The argument on $B_j \setminus \{t\}$ is the same. Its augmented cochain complex is exact because this tree is contractible.

At t: a cone on an eight-cycle. There are nine vertices: four n_{ij} from N, four a_i, b_j from $\widetilde{S}$, and t. The eight branches of $\widetilde{E}$ give edges $n_{ij}a_i, n_{ij}b_j$. They form the outer cycle shown below. The points of P and $\widetilde{T}$ give its eight edges to t. Finally, h_{ij}, v_{ij} give the eight triangles (n_{ij}, a_i, t) and (n_{ij}, b_j, t).



9 vertices, 16 edges, 8 triangles.

The realization is a triangulated disk.

Removing the apex and its faces leaves the outer loop.

A cone is contractible, so its augmented cochain complex is exact too. Exactness on all stalks proves $\mathbb{Q}_Z \xrightarrow{\sim} R\varepsilon_*\mathbb{Q}_{Z_\bullet}$. The same argument works with integral coefficients, as in the book's definition of descent. Together with smoothness and properness, this proves that our cube is a hyperresolution. It also illustrates the general preservation of descent under this construction: [1, Lemma 5.20, Lemma 5.27 and Corollary 5.28] express the same argument through iterated cones.

6.4 Cohomology, weights, and comparison with the intersection cube

We now calculate with the cube just constructed. From (7), the nonzero rows of the spectral sequence (3) are

$s\backslash r$	0	1	2
4	$\mathbb{Q}(-2)^4$	0	0
2	$\mathbb{Q}(-1)^{12}$	$\mathbb{Q}(-1)^8$	0
0	$\mathbb{Q}^9$	$\mathbb{Q}^{16}$	$\mathbb{Q}^8$

(d_1 goes to the right). (8)

The bottom row. Every connected component of every entry has one point above t, so its degree-zero restriction maps are exactly the cochain maps of the disk just drawn. Therefore its cohomology is $\mathbb{Q}$ in column 0 and zero in the other columns. In particular, the eight triangles remove the apparent loop, rather than adding a cohomology class.

The middle row. The four surfaces in N contribute eight ruling classes, and the four branches of $\widetilde{S}$ contribute four curve classes. Write

$$\alpha_{ij} = x_{ij}\ \mathrm{pr}_1^*[\mathrm{pt}] + y_{ij}\ \mathrm{pr}_2^*[\mathrm{pt}] \in H^2(N_{ij}), \qquad u_i \in H^2(A_i), \quad v_j \in H^2(B_j),$$

using the same symbols u_i, v_j for their coefficients. Restriction to $H_{ij} = L_i \times \{s_j\}$ selects x_{ij}, while restriction to $V_{ij} = \{s_i\} \times L_j$ selects y_{ij}. With the ordered indices $0 < 1 < 2$, the alternating differential to $Z_{\{0,1\}} = E$ is

$$d_1 : \mathbb{Q}(-1)^{12} \longrightarrow \mathbb{Q}(-1)^8, \qquad (x, y, u, v) \longmapsto \big(u_i - x_{ij},\ v_j - y_{ij}\big)_{i,j=0,1}.$$

The point corners contribute nothing in degree 2. Each output has its own variable x_{ij} or y_{ij}, so the map is onto. Its kernel is given by $x_{ij} = u_i$, $y_{ij} = v_j$, with the four free parameters u_0, u_1, v_0, v_1. Thus $E_2^{0,2} = \mathbb{Q}(-1)^4$ and $E_2^{1,2} = 0$.

The top row and the MHS. The four degree-four surface classes have no possible target and survive. Hence E_2 is concentrated in column 0. By (4),

$$\boxed{H^0(Z) = \mathbb{Q}(0), \quad H^2(Z) = \mathbb{Q}(-1)^4, \quad H^4(Z) = \mathbb{Q}(-2)^4, \quad H^{\mathrm{odd}}(Z) = 0.}$$

There is only one weight in each nonzero degree. This agrees with the Künneth isomorphism of MHS and the calculation of $H^*(X)$ in Example 1 ([1, Theorem 5.44]).

Why the full intersection cube also works. There is another valid construction. Label the four components $S_{ij} = L_i \times L_j$ by $1 = 00, 2 = 01, 3 = 10, 4 = 11$, and put $W_\varnothing = Z$, $W_I = \bigcap_{\ell \in I} S_\ell$ for every nonempty $I \subseteq \{1, 2, 3, 4\}$, with inclusion maps. All these intersections are smooth: pairs are $\mathbb{P}^1$ or a point, and triples and the quadruple are points. At z, let A_z be the set of components containing it. The stalk complex is the augmented cochain complex of the *full simplex* on A_z, hence is exact. Properness and smoothness then make this a 4-cubical hyperresolution as well.

At t this second construction has a tetrahedron as its fibre realization; our discriminant construction has a disk. They are different cubical diagrams, both with contractible fibres and proper smooth entries. The independence theorem ([1, Theorem 5.33]) ensures that both produce the same canonical MHS on $H^*(Z)$. The 3-cube above additionally shows how the discriminant-square procedure builds the needed gluing data.

7 A punctured nodal curve: gluing and boundary weights together

Let C be the rational nodal cubic and remove $r \geq 1$ distinct smooth points $p_1, \ldots, p_r$. Put

$$U = C \setminus \{p_1, \ldots, p_r\}, \qquad V = \mathbb{P}^1 \setminus D, \qquad D = \{d_1, \ldots, d_r\},$$

where d_i lies over p_i. The two points a, b lying over the node remain in V. The compactified cubical pair has upper corner $(\mathbb{P}^1, D)$, node corner $(\{s\}, \varnothing)$, and intersection corner $(\{a, b\}, \varnothing)$. Its open part is the normalization square

$$\begin{array}{ccc} \{a,b\} & \longrightarrow & V = \mathbb{P}^1 \setminus D \\ \downarrow & & \downarrow \nu \\ \{s\} & \longrightarrow & U \end{array}$$

First compute the smooth open level. The logarithmic de Rham complex is

$$\mathcal{O}_{\mathbb{P}^1} \xrightarrow{d} \Omega^1_{\mathbb{P}^1}(\log D).$$

Specializing the residue/Gysin construction of [1, §§4.2–4.3, Lemma 4.6 and Proposition 4.10] to $(\mathbb{P}^1, D)$ gives the following exact sequence of MHS:

$$0 \longrightarrow H^1(V) \longrightarrow \mathbb{Q}(-1)^r \xrightarrow{\sum} \mathbb{Q}(-1) \longrightarrow H^2(V) \longrightarrow 0. \tag{9}$$

The middle map adds the residues, equivalently the point classes. It is surjective for $r \geq 1$. Therefore

$$H^0(V) = \mathbb{Q}(0), \qquad H^1(V) = \mathbb{Q}(-1)^{r-1}, \qquad H^2(V) = 0.$$

This is also visible in logarithmic forms: their residues at the d_i sum to zero. The node preimages a, b are *not* logarithmic poles.

Then restore the node. The degree-zero map is again $(u, v) \mapsto (u - v, u - v)$. Its weight-zero cokernel contributes the node loop, and the next term in the long exact sequence is $H^1(V)$. Thus

$$\boxed{0 \longrightarrow \mathbb{Q}(0) \longrightarrow H^1(U) \longrightarrow \mathbb{Q}(-1)^{r-1} \longrightarrow 0} \tag{10}$$

is an exact sequence of MHS. Also $H^0(U) = \mathbb{Q}(0)$ and $H^k(U) = 0$ for $k \geq 2$. In degree one,

$$0 = W_{-1} \subset W_0 = W_1 \simeq \mathbb{Q}(0) \subset W_2 = H^1(U),$$

and the only potentially nonzero mixed Hodge numbers are

$$\boxed{h^{0,0}(H^1(U)) = 1, \qquad h^{1,1}(H^1(U)) = r - 1.}$$

For $r = 1$, degree one is pure of weight 0. For $r = 2$, it has both weights 0 and 2, giving our first example with more than one weight.

Equation (10) determines the weight-graded pieces. It does *not* assert that $H^1(U) \simeq \mathbb{Q}(0) \oplus \mathbb{Q}(-1)^{r-1}$ as MHS. The Hodge filtration can retain extension information, as the following period calculation makes explicit.

8 Beyond the Hodge numbers: an explicit extension

Specialize to $r = 2$. Write the punctures of the normalization as c, d, distinct from the node preimages a, b, and choose a projective coordinate z on $\mathbb{P}^1$. The rational function

$$f(z) = \frac{z-c}{z-d}$$

has divisor $c - d$. The logarithmic form

$$\eta = \frac{1}{2\pi i}\, d\log f = \frac{1}{2\pi i}\left(\frac{dz}{z-c} - \frac{dz}{z-d}\right)$$

spans $F^1H^1(U)_{\mathbb{C}}$ via the filtered complex of Section 4. It is regular at a, b.

Choose a path in V from a to b. After the two endpoints are identified in U, it becomes a loop γ. Let λ be a positive small loop around c. These form a rational homology basis: γ records the node gluing and λ records the punctures. Let α, β be the dual cohomology basis, so

$$\alpha(\gamma) = 1, \quad \alpha(\lambda) = 0, \qquad \beta(\gamma) = 0, \quad \beta(\lambda) = 1.$$

The weight-zero subspace is $W_0 = \mathbb{Q}\alpha$. In the total complex, η has zero component on the points, so the edges closing the path through the node contribute zero. The periods are

$$\int_\lambda \eta = 1, \qquad \int_\gamma \eta = \frac{1}{2\pi i}\int_a^b d\log f =: t.$$

Consequently the complete Hodge filtration is

$$\boxed{F^0 = H^1(U)_{\mathbb{C}}, \qquad F^1 = \mathbb{C}(\beta + t\alpha), \qquad F^2 = 0.} \tag{11}$$

The logarithm depends on the chosen path, and

$$t = \frac{1}{2\pi i}\log\left(\frac{(b-c)(a-d)}{(b-d)(a-c)}\right) \pmod{\mathbb{Z}}, \tag{12}$$

with the usual limiting interpretation at infinity. Changing the path changes γ by an integer multiple of λ and changes the dual basis accordingly; the MHS in (11) is unaffected.

A concrete nonsplit case. Choose $a = 0$, $b = \infty$, $c = 1$, $d = 2$. Then the cross-ratio in (12) is 2, and one may take

$$t = \frac{\log 2}{2\pi i}.$$

A rational splitting of (10) would require a rational line complementary to $\mathbb{Q}\alpha$ whose complexification is F^1. By (11) this is possible exactly when $t \in \mathbb{Q}$. Here t is nonzero and purely imaginary, so no such splitting exists. This conclusion is unchanged by its integral ambiguity.

What the example teaches. The mixed Hodge numbers always remain $h^{0,0} = h^{1,1} = 1$, whereas the relative positions of the punctures and node preimages enter the Hodge filtration through (12). Thus the weight-graded vector spaces and their Hodge numbers do not determine the full MHS. The cubical logarithmic complex retains precisely the period information that a list of Betti numbers or Hodge numbers omits.

9 A calculation recipe and a summary table

For a concrete singular variety, the following sequence of questions keeps the geometry, topology, and filtrations aligned.

1. **Find the smooth pieces.** Normalize a curve; in higher dimensions begin with a resolution and its discriminant square. For an open variety, compactify and keep track of the boundary.
2. **Record every necessary incidence.** Include the exceptional or overlap corners and resolve them compatibly. Verify descent; for a finite component diagram, a stalkwise simplex argument can suffice.
3. **Compute ordinary cohomology.** Write the restriction maps, not just the vector-space dimensions. Take kernels and cokernels in the long exact sequence or the rows of $E_1^{r,s}$.
4. **Read weights and Hodge types.** For smooth proper levels, use $\mathrm{Gr}_s^W H^n = E_2^{n-s,s}$. For open levels, use logarithmic pairs, residues, and the total weight shift (5)–(6).
5. **Separate graded data from extensions.** Record $h^{p,q}(H^k)$ degree by degree. If the full MHS is wanted, inspect F and periods rather than declaring a direct sum from the graded pieces.

The table lists the only potentially nonzero mixed Hodge numbers in the worked examples. Unlisted degrees and entries vanish.

Variety	Degree k	$h^{p,q}(H^k)$
$X = L_0 \cup L_1 \subset \mathbb{P}^2$	0	$h^{0,0} = 1$
	2	$h^{1,1} = 2$
Rational nodal cubic C	0	$h^{0,0} = 1$
	1	$h^{0,0} = 1$
	2	$h^{1,1} = 1$
$X \times X$	0	$h^{0,0} = 1$
	2	$h^{1,1} = 4$
	4	$h^{2,2} = 4$
$C \setminus \{p_1, \dots, p_r\}$, $r \geq 1$	0	$h^{0,0} = 1$
	1	$h^{0,0} = 1, \quad h^{1,1} = r - 1$

The signed Hodge character $\sum_k (-1)^k [H^k(X)]$ is useful, but it forgets the cohomological degree and possible cancellations. It cannot replace the degreewise calculation above.

References

[1] C. A. M. Peters and J. H. M. Steenbrink, *Mixed Hodge Structures*, Ergebnisse der Mathematik und ihrer Grenzgebiete, 3. Folge, vol. 52, Springer, 2008. Publisher DOI. Public author manuscript dated 11 January 2007: Radboud Repository.

[2] *Corrections to Peters–Steenbrink, Mixed Hodge Structures.* Correction sheet hosted on C. A. M. Peters's author webpage.

Source note. The numbered locators above were checked in the public author manuscript, whose chapter pagination agrees with the publisher's contents. The construction and conventions follow Chapters 3–5; the existence statement is used with the author's errata, without its uncorrected dimension bounds. The explicit examples are calculated here from those constructions.